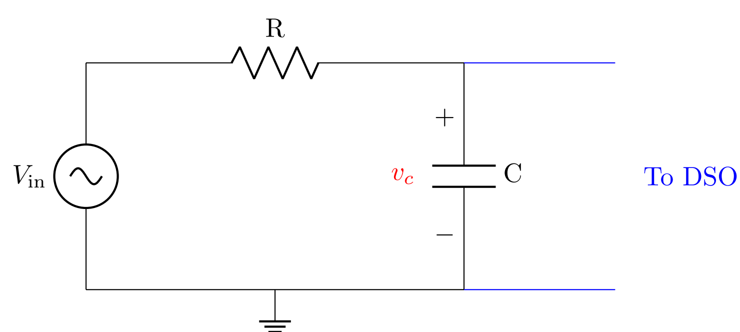
Experiment 7
Filters

Components/Instruments/Meters

- Digital storage oscilloscope (DSO)
- Function generator
- Breadboard
- Resistor
- Capacitor
- Inductor
- Connecting wires

Part 1: Frequency response of low pass filter

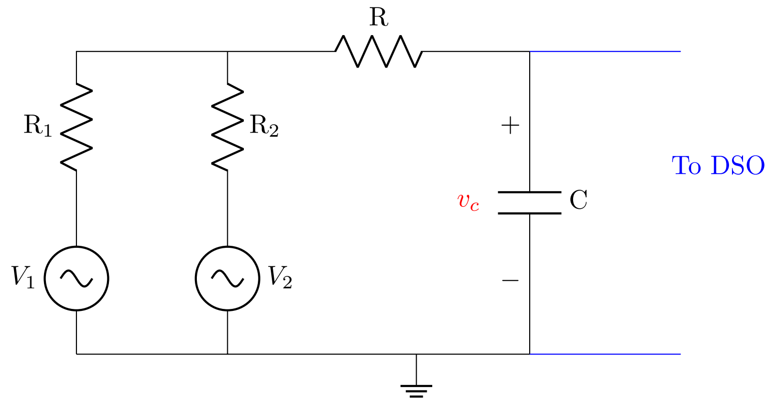
1. Setup a circuit as shown below. Take the value of R as 1 kΩ and C as 4.7 μF.



2. Set the signal generator to sine wave signal (Peak-to-peak value = 2 V, frequency = 10 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the voltage across the capacitor, i.e., v_c.
5. Vary the frequency of the input signal over a range till 10 kHz in 1 kHz increments. Measure the amplitude of the output voltage v_c for different values of frequency.
6. Find the cutoff frequency at which the output voltage across the capacitor is 0.707 times of the input. Compare it with the theoretical value.
7. Plot $\left(\frac{v_c}{V_{\text{in}}}\right)$ vs the frequency.

Part 2: Filter implementation

1. Setup a circuit as shown below. Take the value of R, R_2 and R_3 as 1 kΩ and C as 4.7 μF.



2. Set the function generator as follows:
   Channel-1: Peak-to-peak value = 2 V, frequency = 10 Hz
   Channel-2: Peak-to-peak value = 2 V, frequency = 10 kHz
3. Connect 'Channel-1' of DSO to input terminals of the circuit. (First terminal of "R" and second terminal of "C").
4. Connect 'Channel-2' of DSO to observe the voltage across the capacitor, i.e., v_c.
5. Is the circuit able to filter out the high frequency input? Can you think of a practical application for this circuit?

Optional: How would you modify the circuit if we need to attenuate the high frequency signal more?

Optional: How would you modify the circuit if the low frequency signal needs to be filtered out instead?